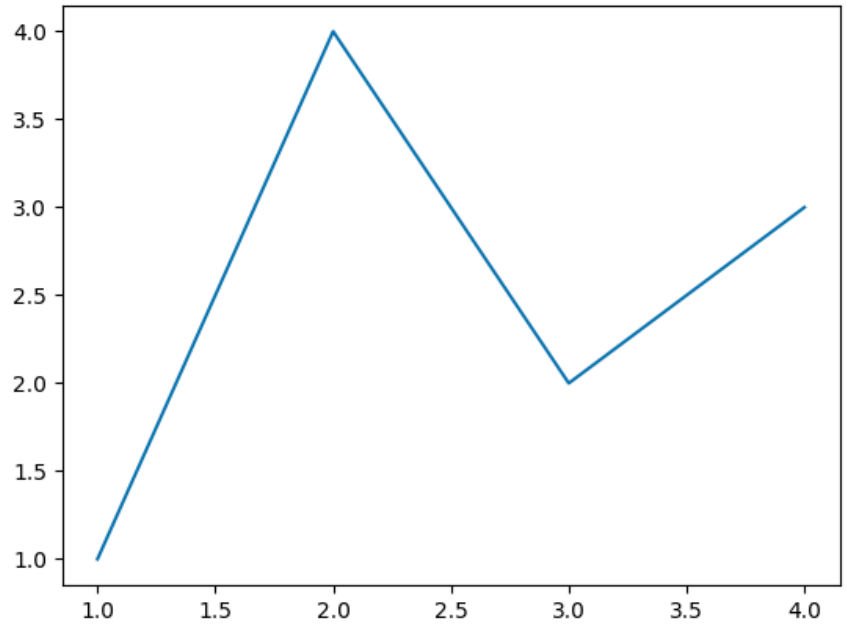
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1. Simple Line Plot

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots()
ax.plot([1, 2, 3, 4], [1, 4, 2, 3])
plt.show()
```



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2. Multiple Line Plot

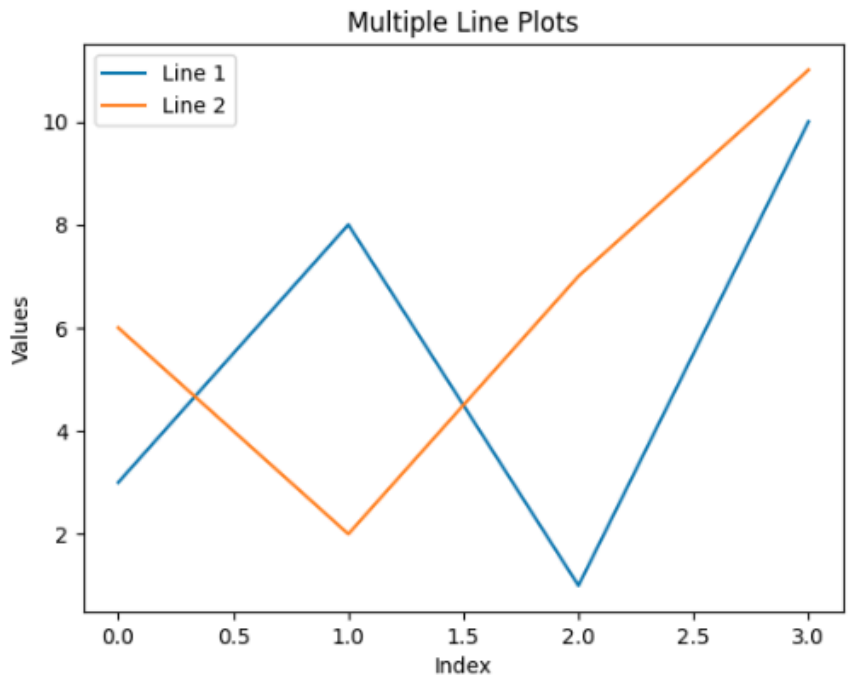
```
import matplotlib.pyplot as plt
import numpy as np

y1 = np.array([3, 8, 1, 10])
y2 = np.array([6, 2, 7, 11])

plt.plot(y1, label="Line 1")
plt.plot(y2, label="Line 2")

plt.title("Multiple Line Plots")
plt.xlabel("Index")
plt.ylabel("Values")
plt.legend()

plt.show()
```



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3. Scatter Plot

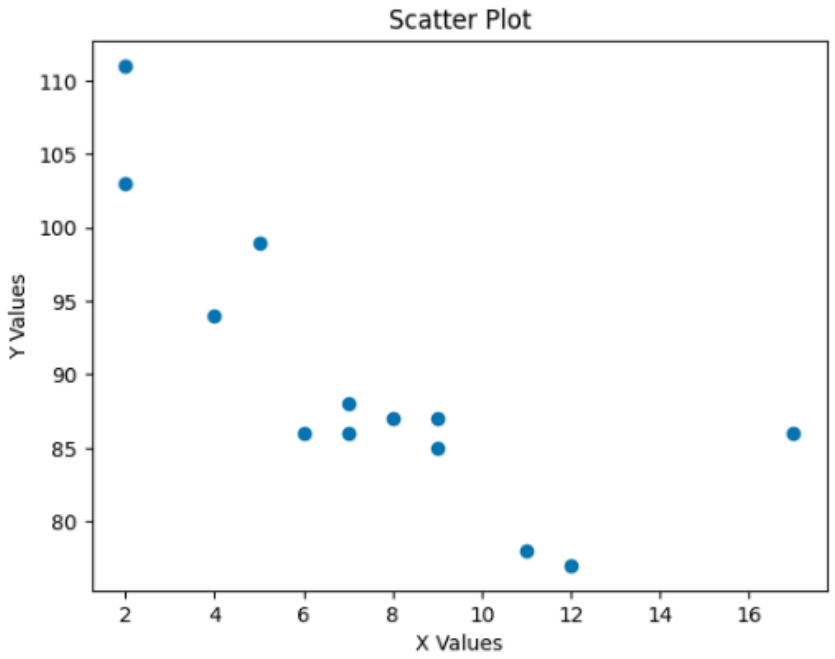
```
import matplotlib.pyplot as plt
import numpy as np

x = np.array([5,7,8,7,2,17,2,9,4,11,12,9,6])
y = np.array([99,86,87,88,111,86,103,87,94,78,77,85,86])

plt.scatter(x, y)

plt.title("Scatter Plot")
plt.xlabel("X Values")
plt.ylabel("Y Values")

plt.show()
```



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4. Random Data Scatter Plot

```
from matplotlib import pyplot as plt
import numpy as np

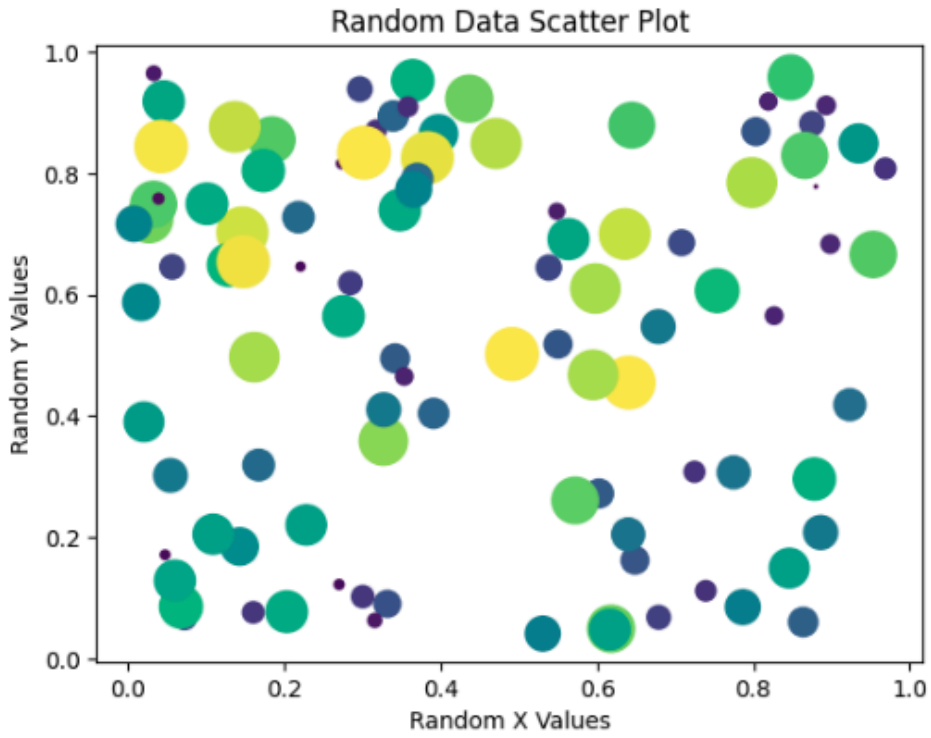
x, y, scale = np.random.rand(3, 100)

fig, ax = plt.subplots()

ax.scatter(x=x, y=y, c=scale, s=np.abs(scale)*500)

ax.set_title("Random Data Scatter Plot")
ax.set_xlabel("Random X Values")
ax.set_ylabel("Random Y Values")

plt.show()
```



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5. Bar Graph

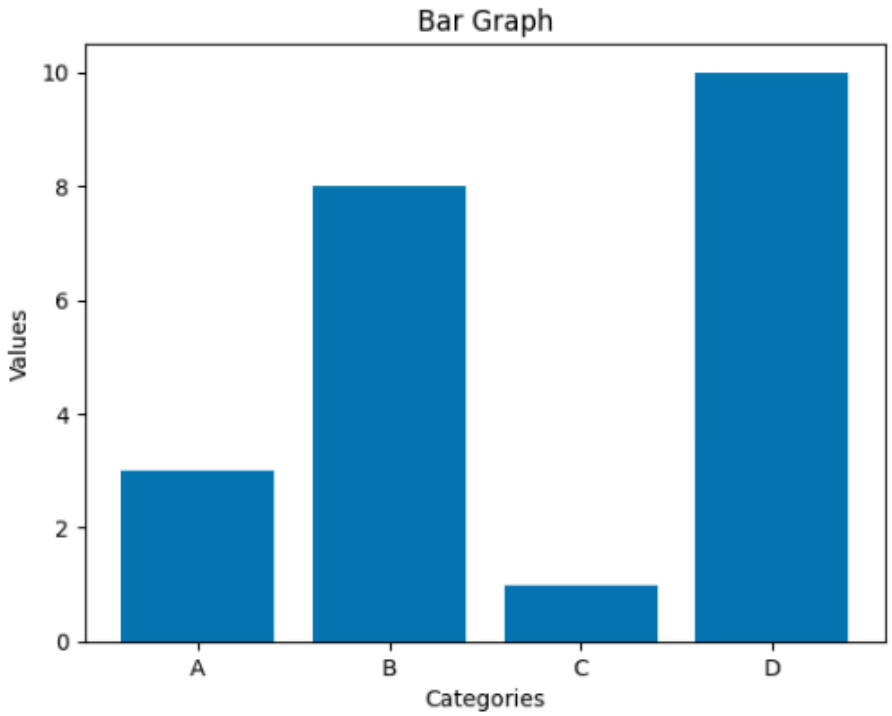
```
import matplotlib.pyplot as plt
import numpy as np

x = np.array(["A", "B", "C", "D"])
y = np.array([3, 8, 1, 10])

plt.bar(x, y)

plt.title("Bar Graph")
plt.xlabel("Categories")
plt.ylabel("Values")

plt.show()
```



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6.

```
import numpy as np

x = np.random.normal(170, 10, 250)

print(x)
```

```
[155.36822553 165.98457207 166.83843661 181.00045253 161.28408639
 164.92988684 165.19760916 187.86349253 152.4753268  180.09235033
 170.8664415  172.08914834 167.52729694 155.48378413 165.20006441
 161.30753646 176.16204725 160.2173558  194.99230499 165.10012206
 174.04615665 170.61906155 170.15936707 165.4185885  150.19794259
 177.27059322 175.27443651 176.00329366 159.74749393 148.77977218
 172.21350249 174.2989457  158.71504646 155.82685726 168.38784923
 164.25807258 171.0735951  171.04447493 162.56649197 158.49709267
 162.95412269 171.12132718 166.97912487 167.48657064 181.83831402
 182.12707506 167.42494197 150.36803154 180.7307077  172.06615514
 159.16091881 166.18627259 164.53979504 168.31340415 173.79877346
 178.16911721 154.98306619 173.50157143 161.36951094 181.17599736
 171.2517434  180.95102906 147.44617675 168.425322   163.68827334
 161.71964328 168.39392344 157.34161559 181.18160964 167.06430336
 168.81729965 171.91347835 165.77820072 162.23847849 169.38337757
 177.9951793  172.17942296 171.82304814 193.93422948 158.54961999
 184.50872837 158.79735126 164.47079899 171.61934916 154.52267314
 162.89797286 179.17389983 190.2745359  177.91604322 190.76505662
 161.70225131 160.7844808  154.68133622 164.50343477 170.98439298
 160.5962122  163.06886746 170.4745806  184.27416184 158.80387259
 168.32008114 161.59731473 173.45754319 165.13757695 163.79465342
 176.83033114 167.41401766 167.85351387 170.16915402 164.3915274
 166.97368491 160.16424155 177.61758994 169.55886608 181.5135546
 163.19975222 175.66274321 150.32355253 164.17029259 160.8748228
 171.34518859 159.66027038 164.38297618 157.10655975 174.41158395
 163.26159984 176.29752275 156.44384339 178.02611095 155.6843442
 172.38961474 196.13731754 193.62567903 192.36068662 164.7009901
 175.66511207 155.53382575 166.41118678 171.18857899 159.06881461
 171.19492096 159.76760755 183.98468224 166.92716051 190.72835366
 171.75741823 185.5590914  175.04479928 175.24875126 166.04624661
 168.90038655 151.93102146 151.62185563 175.29086263 163.80631574
 171.89814455 164.23788741 153.6138477  147.48009441 180.78524886
 173.41932604 168.1639702  184.1004101  161.05053578 170.67873896
 172.37884284 175.04119794 182.17139987 163.9978109  180.02882342
 169.36309453 168.72908783 158.30021002 180.52761773 189.78226664
 163.77808513 170.02513943 173.36219563 185.38849134 158.15543702
 172.89446214 164.02519663 180.81859879 186.23489666 172.96975424
 177.88199556 161.63418919 165.54162224 167.08183383 174.55084271
 158.00028152 167.04762339 182.82724537 194.03121848 185.80140716
 164.86460717 177.28364141 172.61503208 148.44404965 165.12152539
 169.01225804 158.11027561 185.06839237 179.0772548  180.87954563
 176.09421918 174.68469284 172.77037178 169.19392076 183.31163724
 163.82535932 163.98042501 172.61925404 175.95945433 190.01095187
 169.82634143 169.76170234 185.95740994 181.03749212 180.69024524
 170.83740269 166.8564346  176.27404576 184.18024358 150.4213164
 155.80201284 183.13716056 187.42583576 169.65892971 177.11426955
 172.86653136 154.90076803 175.18192859 172.17034729 171.56884092
 162.37003378 162.86481677 165.57675853 172.97122472 166.99467319
 142.01971433 159.33736916 163.35814553 179.38073983 170.15127734
 156.23408226 169.20017619 163.09023435 166.85989403 160.13850927]
```

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7. Histogram of Normal Distribution

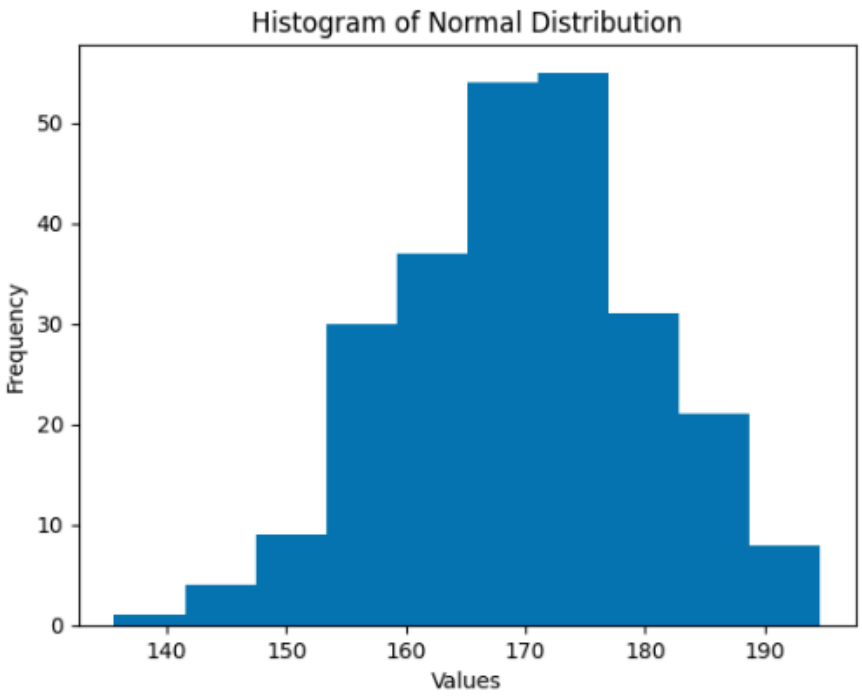
```
import matplotlib.pyplot as plt
import numpy as np

x = np.random.normal(170, 10, 250)

plt.hist(x)

plt.title("Histogram of Normal Distribution")
plt.xlabel("Values")
plt.ylabel("Frequency")

plt.show()
```



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8. Pie Chart

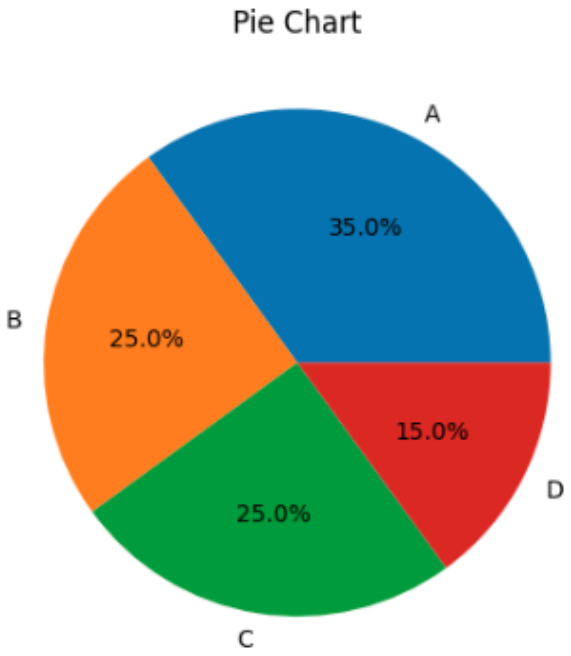
```
import matplotlib.pyplot as plt
import numpy as np

y = np.array([35, 25, 25, 15])

plt.pie(y, labels=["A", "B", "C", "D"], autopct="%1.1f%%")

plt.title("Pie Chart")

plt.show()
```



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9.Monthly Sales and Expenses

```
import pandas as pd
import matplotlib.pyplot as plt

# Read data from CSV file
df = pd.read_csv("sales_data.csv")

# Display the data
print(df)

# Create a line chart
plt.plot(df["Month"], df["Sales"],
         marker="o", label="Sales", color="blue")

plt.plot(df["Month"], df["Expenses"],
         marker="s", label="Expenses", color="red")

# Add chart information
plt.title("Monthly Sales and Expenses")
plt.xlabel("Month")
plt.ylabel("Amount ($)")
plt.legend()
plt.grid(True)

# Display chart
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

```
      Month  Sales  Expenses
0   January  15000      8000
1  February  18000      9000
2     March  22000     11000
3     April  25000     12000
4       May  30000     15000
5      June  28000     14000
```



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10. Passenger Statistics

```
import pandas as pd
import matplotlib.pyplot as plt

# URL of online CSV dataset
url = "https://raw.githubusercontent.com/mwaskom/seaborn-data/master/flights.csv"

# Read data directly from the website
df = pd.read_csv(url)

# Convert year and month into dates
df["Date"] = pd.to_datetime(
    df["year"].astype(str) + "-" + df["month"],
    format="%Y-%B"
)

# Display first 10 records
print(df.head(10))

# Create line chart
plt.figure(figsize=(12, 6))

plt.plot(
    df["Date"],
    df["passengers"],
    color="blue",
    marker="o",
    markersize=3,
    linewidth=1.5
)

# Chart title and labels
plt.title("Airline Passenger Statistics (1949-1960)")
plt.xlabel("Year")
plt.ylabel("Number of Passengers (Thousands)")

# Add grid
plt.grid(True)

# Improve chart layout
plt.tight_layout()

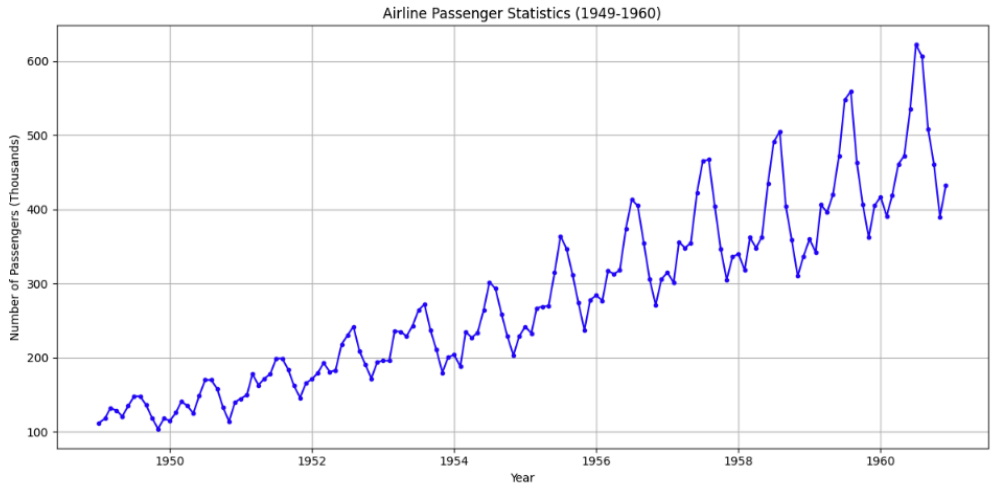
# Display chart
plt.show()
```

```
   year      month  passengers       Date
0  1949    January         112 1949-01-01
1  1949   February         118 1949-02-01
2  1949      March         132 1949-03-01
3  1949      April         129 1949-04-01
4  1949        May         121 1949-05-01
```

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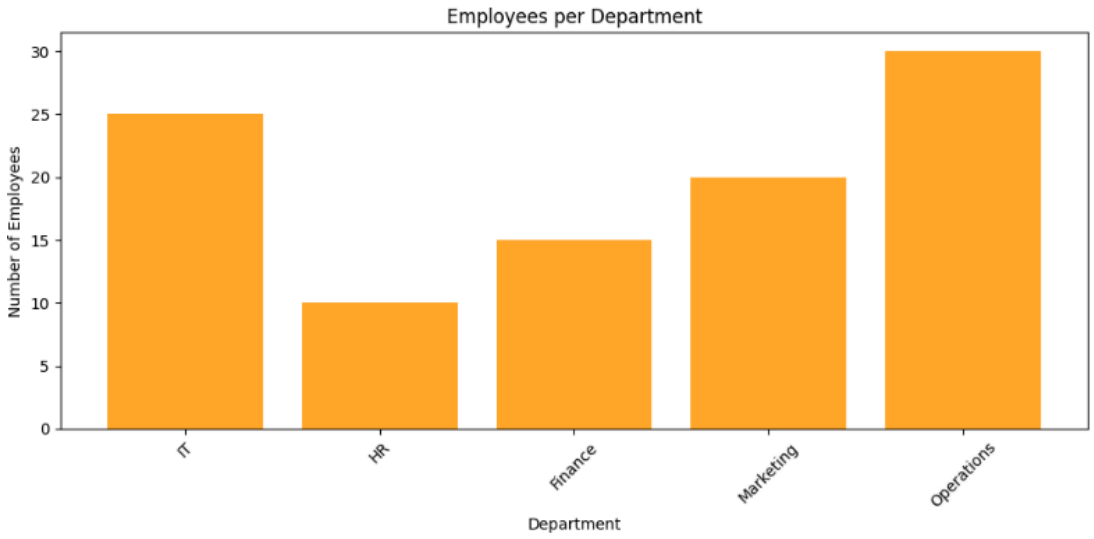
```
5  1949       June         135 1949-06-01
6  1949       July         148 1949-07-01
7  1949     August         148 1949-08-01
8  1949  September         136 1949-09-01
9  1949    October         119 1949-10-01
```



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11. Employees




```
import pandas as pd
import matplotlib.pyplot as plt

# Read Excel file
df = pd.read_excel("company_data.xlsx")

# Display records
print(df.head())

# Create bar chart
plt.figure(figsize=(10, 5))

plt.bar(df["Department"], df["Employees"],
        color="orange")

plt.title("Employees per Department")
plt.xlabel("Department")
plt.ylabel("Number of Employees")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

```
   Department  Employees
0          IT         25
1          HR         10
2     Finance         15
3   Marketing         20
4  Operations         30
```

